

REAL-TIME SPEECH WATERMARKING FOR LIVE VOICE COMMUNICATION: SYNTHETIC SPEECH PROVENANCE AND POST-HOC RECORDING ATTRIBUTION

Yizhu Wen* Huaili Zeng[§] Rui Duan[†] Tianxing Li[§] Nan Zhang[¶] Hanqing Guo*

*Indiana University Bloomington, Department of Computer Science, Bloomington, USA

[§]Michigan State University, Computer Science and Engineering Department, East Lansing, USA

[†]University of Missouri-Kansas City, Department of Computer Science, Kansas City, USA

[¶]Indiana University Bloomington, The Media School, Bloomington, USA

ABSTRACT

The growing use of synthetic speech in live communication creates a need for provenance mechanisms that operate in real time without disrupting speech transmission. Existing neural speech watermarking methods are primarily designed for offline audio and are sensitive to the temporal mismatch and channel distortions introduced by live communication. We propose **Real-Time Speech Watermarking (RT-SW)**, a predictive framework that generates watermarks for upcoming speech from preceding context. RT-SW further incorporates differentiable telephony distortions during training and a TF-Loudness loss to improve robustness and imperceptibility. Experiments on both human and synthesized speech show that RT-SW maintains reliable watermark recovery under phone-call transmission while preserving competitive watermark imperceptibility. These results demonstrate the feasibility of robust real-time speech watermarking for live voice communication and post-hoc provenance verification. Code is available at https://github.com/yizhu-wen/Realtime_WM

Index Terms— Speech Watermark, Real-Time

1. INTRODUCTION

Recent advances in speech generation and voice conversion are bringing synthetic speech into live communication, where speech may be generated and transmitted continuously during an ongoing conversation [1, 2]. This creates provenance requirements for both synthetic and human speech. Speech watermarking has seen widespread use for this purpose in the past few decades. For synthetic speech, a watermark embedded at generation time can indicate that the transmitted voice originated from an AI system and can remain detectable if the conversation is later recorded or retransmitted [3, 4]. For human speech, a watermark inserted into the outgoing speech stream can bind provenance information to the live communication session, allowing a later recording to be attributed to the corresponding source [5, 6].

Advanced neural audio watermarking methods have achieved strong robustness and imperceptibility under com-

mon signal-processing transformations than traditional ones [7]. Methods like WavMark [8] and AudioSeal [3] demonstrate effective watermark embedding under compression, resampling, filtering, and other distortions. However, these methods mainly target offline watermarking, where the full audio is available before embedding, without considering a VoIP telephony channel distortion. In live communication, processing latency causes a watermark generated from the current speech to be embedded into future speech frames, creating a mismatch between the speech used to generate the watermark and the speech that carries it. Our experiments show that this mismatch substantially degrades watermarking effectiveness. On the other hand, prior work demonstrates real-time audio watermarking using traditional methods such as LSB and spread-spectrum embedding [9], including evaluations over VoIP codecs [5]. However, these approaches rely on fixed, hand-crafted embedding and perceptual strategies, and their robustness varies considerably across codecs and signal-processing distortions [10].

To address the *real-time* and *telephony-robustness* challenges, we propose **Real-Time Speech Watermarking (RT-SW)**, a predictive watermarking framework inspired by predictive attacks on automatic speech recognition (ASR) [11]. For *real-time* embedding, RT-SW predicts the watermark for upcoming speech frames using two seconds of preceding speech as the input, with the prediction window continuously advancing as the conversation proceeds. To improve *robustness* to telephony VoIP transmission, RT-SW generates watermark in the time-frequency domain using the Short-Time Fourier Transform (STFT) and introduces a differentiable telephony distortion layer between the watermark embedder and decoder during model training. To improve *imperceptibility*, we incorporate a TF-Loudness perceptual loss [3] which encourages the watermark to be hidden in acoustically busy regions of speech where it is less perceptible to human listeners. Extensive experiments on both human and synthesized speech demonstrate its robustness under telephony distortions while maintaining high imperceptibility. The contributions can be summarized as follow: 1) We propose

RT-SW, a predictive watermarking framework that generates watermarks for live speech; 2) We integrate a telephony-aware training pipeline with a TF-Loudness loss to improve robustness with high imperceptibility; 3) Our evaluation both human and synthesized speech demonstrates strong robustness under telephony distortion and high imperceptibility compared with four offline watermarking methods.

2. RELATED WORK

Neural Speech Watermarking. Recent neural audio watermarking methods have improved robustness and imperceptibility through jointly learned embedding and extraction. WavMark [8] introduces robust multi-bit watermarking under common signal-processing distortions, while Timbre Watermarking [4] targets voice-cloning attacks through distortion-aware speech watermarking. MaskMark [12] extends robustness to real and synthetic speech. AudioSeal [3] and SilentCipher [13] further improve perceptual quality and robustness through psychoacoustic modeling. More recent methods, such as DiscreteWM [14] explores discrete representations for robust watermark recovery. However, they assume that the speech to be watermarked is available offline, rather than being transmitted during live communication.

Real-Time Watermarking. Prior work has explored real-time audio watermarking primarily through traditional signal-processing methods or exploiting dedicated hardware. Citaker and Gul [5] process speech in fixed-size frames and embed source-authentication information using lightweight operations suitable for real-time VoIP transmission. Azzaz *et al.* [10] achieve real-time speech watermarking through sample-level LSB embedding implemented on dedicated FPGA hardware, enabling concurrent speech acquisition, watermarking, and transmission. Santin-Cruz and Dolecek [9] minimize embedding latency by using additive time-domain spread-spectrum watermarking with small audio buffers. In contrast, RT-SW utilizes a neural watermark generator with a simulated telephony distortions to generate content-aware and robust to communication-channel transformations.

3. PRELIMINARY STUDY

To evaluate whether SOTA offline watermarking methods can be directly applied to real-time speech, we test four open-source methods on 50 LibriSpeech development clips. We then extract each watermark, shift it forward in time, and add it to the clean audio to simulate applying a watermark generated from past speech to future frames. As shown in Fig. 1, WavMark, AudioSeal, and SilentCipher degrade substantially under temporal shifts, while Timbre remains above 89%. These results show that most offline methods are sensitive to the temporal mismatch introduced by real-time deployment. Although Timbre is more robust, its weaker robustness to phone call distortion and low imperceptibility limits its suitability for live communication, as discussed in Sec. 5.2.

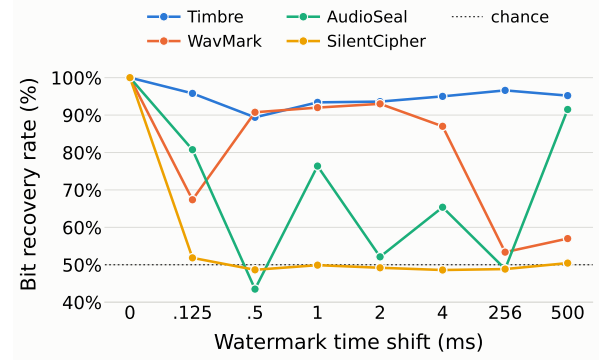


Fig. 1: Bit recovery rate of four offline speech watermarking methods under temporal shift.

4. METHOD

Figure 2 shows an overview of RT-SW. It consists of three components: a predictive real-time *watermark encoder*, a *watermark decoder*, and a *differentiable telephony distortion layer* that emulates the effects during online communication. Below, we describe each component in detail.

4.1. Real-time Watermark Encoder

Predictive Watermark. As shown on the left of Fig. 2, let x^t denote the speech observed at time t . Given x^t and a predefined n -bit message w , the encoder takes δ seconds to generate a watermark for upcoming speech. Inspired by predictive adversarial attacks on ASR [11], RT-SW learns a watermark $a_w^{t+\delta+\alpha}$ from the preceding speech context, such that it remains effective over possible variations in the upcoming utterance. Once generated, the watermark is additively added to the future speech. As the conversation proceeds, this prediction window advances continuously, enabling watermark embedding without delaying the speech stream. We formulate predictive watermark generation as

$$a_w^{t+\delta+\alpha} = f_\theta(x^t, w), \quad (1)$$

where f_θ is the predictive watermark generator parameterized by θ , δ is the inference delay, and α denotes the duration of the predicted watermark. In our experiment, we use $t = 2$, $\delta = \alpha = 0.5$ s. For simplicity, we use x for input speech and a_w for the predicted watermark.

RT-SW first transforms the input speech into the time-frequency domain. Given a single-channel waveform $x \in \mathbb{R}^T$, we apply STFT and encode the resulting coefficients using an feature encoder $\mathbf{EN}_x$:

$$f_x = \mathbf{EN}_x(\text{STFT}(x)), \quad \text{STFT}(x) \in \mathbb{C}^{F \times N} \quad (2)$$

where F is the frequency bins and N is the time frames.

Redundant Frequency Distribution. Prior work [4, 7] shows that repeating watermark information across time dimension improves robustness by allowing watermark recovery from partially preserved audio. We extend this redundancy to frequency dimensions. The n -bit message w is first encoded into a watermark feature:

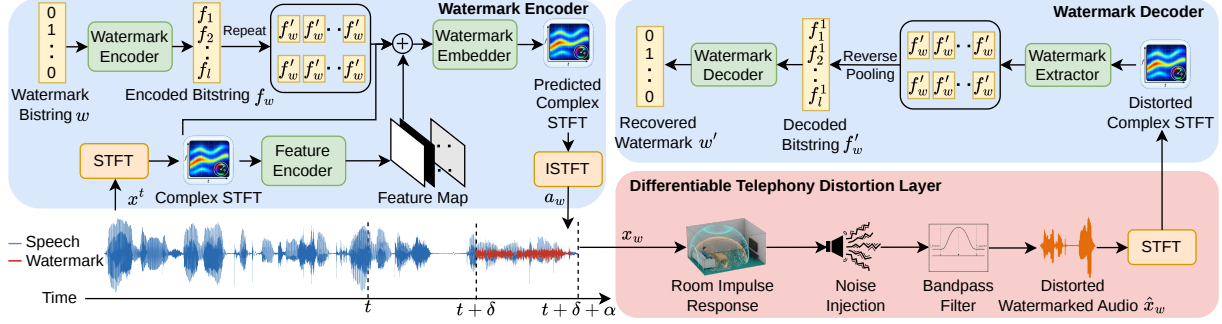


Fig. 2: Proposed real-time audio watermarking pipeline. The watermark generator embeds encoded bitstrings into speech, while the watermark detector reconstructs them after potential transmission distortions.

$$f_w = \mathbf{EN}_w(w), \quad f_w \in \mathbb{R}^{64 \times 1 \times F/2}. \quad (3)$$

We then repeat f_w across all T time frames and twice along the frequency axis to cover the full frequency bands of f_x . Temporal repetition makes recovery less sensitive to cropping and temporal shifts, while frequency repetition improves robustness when part of the spectrum is suppressed. We denote this operation as $\text{Repeat}(\cdot)$.

To preserve the host spectrum, we concatenate f_x with the original STFT results and the repeated watermark feature:

$$f_{\text{full}} = \mathbf{Concat}(f_x, \mathbf{STFT}, \text{Repeat}(f_w, T, 2)), \quad (4)$$

The watermark embedder $\mathbf{EM}$ predicts the watermark STFT coefficient, which is converted back to the waveform domain using inverse STFT ($\mathbf{ISTFT}$) and added to the speech:

$$\begin{aligned} a_w &= \mathbf{ISTFT}(\mathbf{EM}(f_{\text{full}})), \\ x_w &= x + a_w. \end{aligned} \quad (5)$$

4.2. Watermark Decoder

Given watermarked speech x_w , the decoder first uses the watermark extractor $\mathbf{EXT}$ to extract watermark features from its STFT coefficients. Since the watermark is repeated across all time frames and both frequency halves during embedding, reverse-pooling process first averages the extracted features over all N time frames within each frequency half, and then averages the two frequency features before feeding it to the watermark decoder $\mathbf{DE}$ to get the recovered the message w' :

$$w' = \mathbf{DE} \left(\frac{1}{2} \sum_{b=1}^2 \frac{1}{N} \sum_{\tau=1}^N \mathbf{EXT}(\mathbf{STFT}(x_w))[b][:, \tau] \right), \quad (6)$$

Here, $\mathbf{EXT}$ extracts the watermark representation, b indexes the two frequency halves, N denotes the number of frames.

4.3. Training Pipeline

4.3.1. Differentiable Telephony Distortion Layer

To improve robustness to real-world communication channels, we insert a differentiable distortion layer $\mathbf{DP}$ between

the watermark encoder and decoder during training. It simulates common distortions encountered in voice communication, including room reverberation [15], additive Gaussian noise, and telephone-band frequency limitation [16]. Given watermarked speech x_w , the distorted signal is modeled as:

$$\hat{x}_w = \mathbf{DP}(x_w) = \mathbf{BP}(x_w * h + \epsilon), \quad (7)$$

where h denotes a random room impulse response, ϵ denotes additive Gaussian noise, and $\mathbf{BP}$ represents telephone-band filtering. By training the watermark encoder and decoder with differentiable distortions, RT-SW learns robust watermarks that remain recoverable after the telephony channels.

4.3.2. Losses

Perceptual Loss. To preserve speech quality, following Timbre [4], we combine two perceptual losses. An $L2$ loss $\mathcal{L}_w$ limits the magnitude of the watermark perturbation by keeping the watermarked waveform close to the original speech. In contrast, an *adversarial loss* $\mathcal{L}_{\text{adv}}$ encourages the watermarked speech to remain similar to unwatermarked speech.

TF-Loudness Loss. To improve imperceptibility, we adopt the TF-Loudness loss from AudioCraft¹. We divide the speech and watermark into local time-frequency regions and measure their relative loudness:

$$\mathcal{L}_{\text{loud}}^{b,t} = \text{Loudness}(a_{b,t}) - \text{Loudness}(x_{b,t}), \quad (8)$$

where b and t index the frequency band and temporal segment. Minimizing the loudness difference encourages the watermark to remain less audible than the surrounding speech.

Decoding Loss. To ensure message recovery, we use a decoding loss $\mathcal{L}_{\text{dec}}$ based on the $L2$ loss between the embedded message w and the recovered message w' .

The overall training loss is:

$$\mathcal{L} = \mathcal{L}_w + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{loud}}^{b,t} + \mathcal{L}_{\text{dec}}. \quad (9)$$

5. EVALUATION

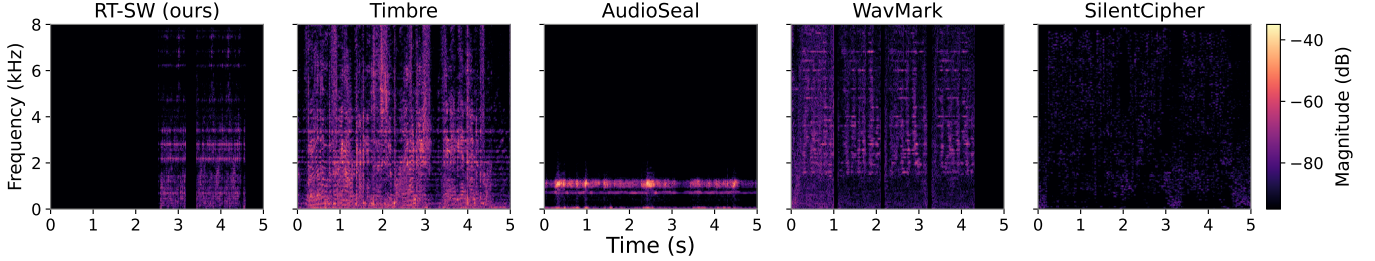
5.1. Experimental Setup

Datasets. We train RT-SW on the LibriSpeech [17] and evaluate on both human and synthesized speech. For human speech, we use the LibriSpeech dev split and randomly

¹<https://github.com/facebookresearch/audiocraft.git>

Table 1: Decoding evaluation results under different audio distortions. All baselines process the audio offline.

Distortion	RT-SW (Ours)			AudioSeal			WavMark			Timbre			SilentCipher		
	Acc.↑	TPR↑/FPR↓	AUC↑	Acc.↑	TPR↑/FPR↓	AUC↑	Acc.↑	TPR↑/FPR↓	AUC↑	Acc.↑	TPR↑/FPR↓	AUC↑	Acc.↑	TPR↑/FPR↓	AUC↑
No Distortion	0.976	0.934/0.009	0.992	1.000	1.000/0.011	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.010	1.000	0.999	0.997/0.011	0.999
Median Filter	0.943	0.839/0.009	0.978	1.000	1.000/0.010	1.000	0.999	0.999/0.010	1.000	1.000	1.000/0.010	0.999	0.982	0.969/0.010	0.984
Low-pass 4k	0.938	0.824/0.010	0.974	1.000	1.000/0.010	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.008	1.000	0.889	0.803/0.008	0.899
High-pass 500	0.968	0.907/0.011	0.988	1.000	1.000/0.011	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.009	1.000	0.965	0.939/0.011	0.969
Compression	0.969	0.915/0.010	0.990	1.000	1.000/0.011	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.009	0.999	0.998	0.996/0.010	0.998
Noise Suppression	0.929	0.805/0.011	0.964	0.998	0.999/0.011	1.000	0.980	0.964/0.010	0.982	0.998	0.996/0.010	0.998	0.877	0.801/0.011	0.900
MP3 64 kbps	0.975	0.932/0.008	0.993	1.000	1.000/0.009	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.011	0.999	0.964	0.938/0.010	0.969
AAC 64 kbps	0.981	0.947/0.011	0.994	1.000	1.000/0.010	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.010	1.000	0.868	0.770/0.012	0.883
Ogg Vorbis	0.978	0.939/0.010	0.993	1.000	1.000/0.010	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.012	0.999	0.963	0.935/0.011	0.967
Sample Suppression	0.978	0.939/0.010	0.994	1.000	1.000/0.009	1.000	1.000	1.000/0.010	1.000	1.000	1.000/0.012	0.999	0.893	0.813/0.009	0.908
Telephony Distortion	0.906	0.734/0.011	0.959	0.535	0.018/0.010	0.582	0.500	0.010/0.010	0.500	0.780	0.299/0.010	0.908	0.499	0.006/0.006	0.498

**Fig. 3:** Comparison of watermark spectrograms across five speech watermarking methods.

sample 2,700 clips from LJSpeech [18]. For synthesized speech, we generate clips from LibriSpeech utterances under HiFi-GAN [19] vocoder resynthesis (2,000 clips) and cross-speaker voice cloning with F5-TTS [20] (2,000 clips).

Metrics. We evaluate message recovery using bit recovery accuracy (Acc), defined as the fraction of correctly recovered watermark bits. For watermark detection, we use the per-clip Acc. that exceeds a threshold. By varying this threshold, we compute the true-positive rate (TPR), false-positive rate (FPR), and area under the ROC curve (AUC). For reporting, we select the threshold that yields approximately 1% FPR and report the corresponding TPR. To evaluate imperceptibility, we report signal-to-noise ratio (SNR) and Perceptual Evaluation of Speech Quality (PESQ).

5.2. Main Results

Robustness. Table 1 compares RT-SW with four SOTA watermarking methods under different audio distortions. Without distortion, all methods perform strongly, with RT-SW reaching 97.4% Acc and an AUC of 0.992. RT-SW also remains robust to median filtering, frequency filtering, audio codec, compression, and noise suppression, etc., with AUC values consistently above 0.95. Under telephony distortion, RT-SW achieves 90.6% Acc and an AUC of 0.956, compared with 78.2%/0.908 for Timbre, 53.5%/0.582 for AudioSeal, 50.0%/0.500 for WavMark, and 50.0%/0.498 for SilentCipher. RT-SW has the highest TPR (73.5%), substantially higher than all other methods. These results show that telephony-aware training improves watermark robustness under live communication channels while maintaining competitive performance under standard audio processing.

Imperceptibility. Table 2 shows that RT-SW achieves strong perceptual quality, with an SNR of 40.22 dB and a PESQ

score of 4.163. Its SNR and PESQ are substantially higher than the three baselines. It is second to SilentCipher, which trades robustness for higher imperceptibility. Figure 3 shows the distribution of watermark energy in the spectrogram. Compared with the more broadly distributed patterns of Timbre and WavMark, RT-SW concentrates its watermark energy in selected frequency regions while maintaining low energy levels across the remaining spectrum. This aligns with the TF-Loudness loss objective, which encourages watermarking in acoustically dominant regions where it is less audible.

Table 2: Imperceptibility metrics of the 4 watermarking methods.

Method	SNR (dB) ↑	PESQ ↑
RT-SW (ours)	40.22	4.163
AudioSeal	27.26	4.426
WavMark	36.61	4.154
Timbre	27.47	3.697
SilentCipher	49.01	4.590

6. CONCLUSION

This work shows that watermarking methods designed for offline speech do not directly translate to live voice communication, where prediction latency and telephony distortions introduce additional challenges. We present RT-SW, which predicts watermarks for upcoming speech and jointly optimizes robustness to communication-channel distortions and perceptual quality. Our results demonstrate that RT-SW maintains reliable watermark detection under phone-call transmission while preserving competitive watermark imperceptibility. More broadly, these findings suggest that real-time speech provenance requires watermarking systems to be designed explicitly for streaming and communication-channel conditions.

7. REFERENCES

- [1] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour, “Moshi: a speech-text foundation model for real-time dialogue,” *arXiv preprint arXiv:2410.00037*, 2024.
- [2] Yisi Liu, Chenyang Wang, Hanjo Kim, Raniya Khan, and Gopala Anumanchipalli, “Rt-vc: Real-time zero-shot voice conversion with speech articulatory coding,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 2025, pp. 385–393.
- [3] Robin San Roman, Pierre Fernandez, Alexandre Défossez, Teddy Furon, Tuan Tran, and Hady Elsahar, “Proactive detection of voice cloning with localized watermarking,” *arXiv preprint arXiv:2401.17264*, 2024.
- [4] Chang Liu, Jie Zhang, Tianwei Zhang, Xi Yang, Weiming Zhang, and Nenghai Yu, “Detecting voice cloning attacks via timbre watermarking,” Accepted by NDSS 2024.
- [5] Fusun Çıtak Er and Ensar Gul, “Comparison of digital audio watermarking techniques for the security of voip communications,” in *2011 7th International Conference on Information Assurance and Security (IAS)*. IEEE, 2011, pp. 13–18.
- [6] Marcos Faundez-Zanuy, Jose J Lucena-Molina, and Martin Hagmüller, “Speech watermarking: an approach for the forensic analysis of digital telephonic recordings,” *Journal of forensic sciences*, vol. 55, no. 4, pp. 1080–1087, 2010.
- [7] Yizhu Wen, Ashwin Innuganti, Aaron Bien Ramos, Hanqing Guo, and Qiben Yan, “Sok: How robust is audio watermarking in generative ai models?,” *arXiv preprint arXiv:2503.19176*, 2025.
- [8] Guangyu Chen, Yu Wu, Shujie Liu, Tao Liu, Xiaoyong Du, and Furu Wei, “Wavmark: Watermarking for audio generation,” *arXiv preprint arXiv:2308.12770*, 2023.
- [9] Carlos Jair Santin-Cruz and Gordana Jovanovic Dolecek, “Audio watermarking system in real-time applications,” in *Informatics*. MDPI, 2024, vol. 12, p. 1.
- [10] Mohamed Salah Azzaz, Redouane Kaibou, and Abdeldjalil Smahi, “Fpga implementation using novel co-design approach of real-time speech chaos based cryptowatermarking prototype,” in *2023 International Conference on Advances in Electronics, Control and Communication Systems (ICAECCS)*. IEEE, 2023, pp. 1–6.
- [11] Mia Chiquier, Chengzhi Mao, and Carl Vondrick, “Real-time neural voice camouflage,” *arXiv preprint arXiv:2112.07076*, 2021.
- [12] Patrick O’Reilly, Zeyu Jin, Jiaqi Su, and Bryan Pardo, “Maskmark: Robust neuralwatermarking for real and synthetic speech,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 4650–4654.
- [13] Mayank Kumar Singh, Naoya Takahashi, Weihsiang Liao, and Yuki Mitsufuji, “SilentCipher: Deep Audio Watermarking,” in *Interspeech 2024*, 2024, pp. 2235–2239.
- [14] Shengpeng Ji, Ziyue Jiang, Jialong Zuo, Minghui Fang, Yifu Chen, Tao Jin, and Zhou Zhao, “Speech watermarking with discrete intermediate representations,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, vol. 39, pp. 24239–24247.
- [15] Kalle J Palomäki, Guy J Brown, and Jon P Barker, “Techniques for handling convolutional distortion with-missing data’automatic speech recognition,” *Speech Communication*, vol. 43, no. 1-2, pp. 123–142, 2004.
- [16] International Telecommunication Union, “Transmission characteristics for telephone band (300–3400 hz) digital loudspeaking and hands-free telephony terminals,” ITU-T Recommendation P.342, International Telecommunication Union, May 2000.
- [17] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur, “Librispeech: an asr corpus based on public domain audio books,” in *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.
- [18] Keith Ito and Linda Johnson, “The lj speech dataset,” <https://keithito.com/LJ-Speech-Dataset/>, 2017.
- [19] Jaehyeon Kong et al., “Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis,” in *NeurIPS*, 2020.
- [20] Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, Jian Zhao, Kai Yu, and Xie Chen, “F5-TTS: A fairytaler that fakes fluent and faithful speech with flow matching,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria, 2025, pp. 6255–6271, Association for Computational Linguistics.